RAJALAKSHMI ENGINEERING COLLEGE



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| **Started on** | Monday, 31 August 2026, 7:55 PM |
| **State** | Finished |
| **Completed on** | Monday, 31 August 2026, 7:56 PM |
| **Time taken** | 1 min 6 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

**Output Format**

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int f(int a[],int l,int r,int x)
{
    if(l>r) return -1;
    int m=(l+r)/2;
    if(a[m]==x) return m;
    return a[m]<x ? f(a,m+1,r,x) : f(a,l,m-1,x);
}

int main()
{
    int n,x,i,j;
    scanf("%d",&n);
    int a[n];
    for(i=0;i<n;i++) scanf("%d",&a[i]);
    scanf("%d",&x);

    for(i=0;i<n;i++)
    {
        j=f(a,0,n-1,x-a[i]);
        if(j>=0 && j!=i)
        {
            printf("%d\n%d",a[i],a[j]);
            return 0;
        }
    }
    printf("No");
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 4<br>2<br>4<br>8<br>10<br>14 | 4<br>10 | 4<br>10 | ✔ |

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>2<br>4<br>6<br>8<br>10<br>100 | No | No | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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